STACKS, DISCRIMINANTS: LEMMA-DISCRIMINANT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 755-759 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 761-787, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_discriminant.json (pb=1303, stmt 755-759, proof 761-787).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00JB: sources/stacks-discriminant/pinned-original-source.tex, source lines 787–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00U3: sources/stacks-discriminant/pinned-original-source.tex, source lines 787–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00J6: sources/stacks-discriminant/pinned-original-source.tex, source lines 787–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00EU: sources/stacks-discriminant/pinned-original-source.tex, source lines 787–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BIL: sources/stacks-discriminant/pinned-original-source.tex, source lines 787–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GM: sources/stacks-discriminant/pinned-original-source.tex, source lines 787–787. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\pi : X \to Y$ be a morphism of schemes which is finite locally free. Then π is étale if and only if its discriminant is empty.*

Proof. By Morphisms, Lemma 02GM it suffices to check that the fibres of π are étale. Since the construction of the trace pairing commutes with base change we reduce to the following question: Let k be a field and let A be a finite dimensional k-algebra. Show that A is étale over k if and only if the trace pairing $Q_{A/k} : A \times A \to k$, $(a, b) \mapsto \text{Trace}_{A/k}(ab)$ is nondegenerate.

Assume $Q_{A/k}$ is nondegenerate. If $a \in A$ is a nilpotent element, then ab is nilpotent for all $b \in A$ and we conclude that $Q_{A/k}(a, -)$ is identically zero. Hence A is reduced. Then we can write $A = K_1 \times \ldots \times K_n$ as a product where each K_i is a field (see Algebra, Lemmas 00J6, 00JB, and 00EU). In this case the quadratic space $(A, Q_{A/k})$ is the orthogonal direct sum of the spaces $(K_i, Q_{K_i/k})$. It follows from Fields, Lemma 0BIL that each K_i is separable over k. This means that A is étale over k by Algebra, Lemma 00U3. The converse is proved by reading the argument backwards. □